

VCS Methodology

VMR0015

REVISION TO AMS-III.AV.: LOW GREENHOUSE GAS EMITTING SAFE DRINKING WATER PRODUCTION SYSTEMS

Version 1.0

31 October 2025

Sectoral Scope 3: Energy Demand

The original CDM methodology *AMS-III.AV. Low Greenhouse Gas Emitting Safe Drinking Water Production Systems*, v1.0 was adopted on 15 April 2011. It has been further revised over time. Version 9.0 was adopted on 12 June 2025. This methodology revision must be used with the most recent version of *AMS-III.AV.* available on the CDM website.

Version 1.0 of this methodology revision was developed by Verra.

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1 SUMMARY DESCRIPTION

Additionality, Crediting Method, and Mitigation Outcome	
Additionality	Activity Method
Crediting Baseline	Project Method
Mitigation Outcome	Reductions

The CDM methodology *AMS-III.AV. Low Greenhouse Gas Emitting Safe Drinking Water Production Systems* applies to project activities that introduce low greenhouse gas (GHG) emitting water purification systems to provide safe drinking water and displace water boiling using non-renewable biomass or fossil fuels.

This methodology revision *VMR0015* includes the following updates:

- Introduction of an updated approach to determine the fraction of non-renewable biomass¹
- The requirement to set at validation the leakage adjustment factor to account for leakage related to the use of non-renewable woody biomass saved by the project activity
- Updated emission factor for non-renewable woody biomass (for both CO₂ and non-CO₂ components) and fossil fuels¹
- Added requirement related to the assessment of double counting of emission reductions with REDD+ project(s) and jurisdictional REDD+ program(s)¹
- Added requirements related to adjusting the baseline level of the residence/institution included in the project to account for effects of interacting technologies
- Added requirements related to the compilation and presentation of relevant data for each distributed device

VMR0015 must be used with the most recent version of *AMS-III.AV*. The procedures and requirements of *AMS-III.AV* must be applied unless indicated otherwise in this document.

¹ Adapted from VCS methodology *VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0*

2 SOURCES

This methodology revision applies to the CDM small-scale methodology *AMS-III.AV*. Project proponents must use this methodology revision in conjunction with the most recent version of *AMS-III.AV*.

This methodology uses the following methodologies and tools:

- VCS tool *VT0010 Emissions from Electricity Consumption and Generation*
- VCS tool *VT0008 Additionality Assessment*
- *CDM TOOL03 Tool to Calculate Project or Leakage CO₂ Emissions from Fossil Fuel Combustion*
- *CDM TOOL19 Demonstration of Additionality of Microscale Project Activities*
- *CDM TOOL21 Demonstration of Additionality of Small-scale Project Activities*
- *CDM TOOL30 Calculation of the Fraction of Non-renewable Biomass*
- *CDM AMS-I.E Switch from Non-renewable Biomass for Thermal Applications by the User*
- *CDM Standard: Sampling and Surveys for CDM Project Activities and Programmes of Activities*

3 DEFINITIONS

The definitions in *AMS-III.AV*. and the most recent version of the *VCS Program Definitions* apply to this methodology.

4 APPLICABILITY CONDITIONS

All applicability conditions in the most recent version of *AMS-III.AV*. must be met in addition to the following condition:

- The risk of double counting of the reduced emissions from non-renewable biomass is assessed on a national basis by evaluating at validation and crediting period renewal whether there are REDD+ projects or jurisdictional REDD+ programs whose project boundaries overlap with the expected fuel source area (in the business-as-usual scenario). The project proponent must report on the findings of this assessment in the project description of projects applying *VMR0015*.

5 PROJECT BOUNDARY

The project boundary must be determined following the procedure provided in the most recent version of *AMS-III.AV*.

6 BASELINE SCENARIO

The baseline scenario must be determined following the procedure provided in the most recent version of *AMS-III.AV*.

7 ADDITIONALITY

Paragraph 12 of *AMS-III.AV* must be replaced by the following:

- 12) The additionality of a project activity must be demonstrated by applying the most recent version of one of the following tools appropriate to the project scale:²
 - a) *VT0008 Additionality Assessment* (for any projects)
 - b) *CDM TOOL21 Demonstration of Additionality of Small-scale Project Activities* (for projects with annual GHG emission reductions of up to 60 kt CO₂e in any year of the crediting period)
 - c) *CDM TOOL19 Demonstration of Additionality of Microscale Project Activities* (for projects with annual GHG emission reductions of up to 20 kt CO₂e in any year of the crediting period)

² Project proponents may demonstrate additionality by applying a tool indicated for use at a larger scale than the project activity, but not one indicated for use at a smaller scale than the project activity. For example, a project with 50 kt CO₂e annual reductions may apply *TOOL21* or *VT0008*, but not *TOOL19*.

8 QUANTIFICATION OF REDUCTIONS AND REMOVALS

8.1 Baseline Emissions

Baseline emissions must be determined following the procedures provided in the most recent version of AMS-III.AV., replacing the equations as indicated in the following:

The following equation replaces Equation 1 of AMS-III.AV.:

$$BE_y = QPW_y \times m \times X_{boil} \times SEC \times \sum_i BL_{fuel,i} \times \left((EF_{b,k,CO_2} \times f_{NRB,i,y}) + EF_{b,k,nonCO_2} \right) \times 10^{-9} \times Adj_{LE} \quad (1)$$

Where:

BE_y	= Baseline emissions in year y (t CO ₂ e)
QPW_y	= Total quantity of water purified by the project in year y (L)
m	= Fraction of functional appliances that are providing safe drinking water (%)
X_{boil}	= Fraction of the population served by the project activity for which the common practice of water treatment is or would have been water boiling; determined ex ante through surveys (dimensionless)
SEC	= Specific energy consumption required to boil one liter of water (kJ/L)
$BL_{fuel,i}$	= Proportion of baseline fuel type i (non-renewable biomass and/or fossil fuels) used in the absence of the project activity (fraction)
EF_{b,k,CO_2}	= CO ₂ emission factor for fuel used by baseline device type k in the baseline scenario (t CO ₂ /TJ)
$f_{NRB,i,y}$	= Fraction of non-renewable fuel type i used in the absence of the project activity in year y ; equal to 1 where the baseline fuel is fossil fuel (dimensionless)
$EF_{b,k,nonCO_2}$	= Non-CO ₂ emission factor for fuel used by baseline device type k in the baseline scenario (t CO ₂ e/TJ)
Adj_{LE}	= Adjustment factor to account for leakage related to non-renewable woody biomass saved by the project activity (dimensionless)

The interaction between a Safe Drinking Water (SDW) system/device and improved thermal energy generation technologies (like improved cookstoves) in households/facilities must be taken into account when determining the level of baseline emissions. η_{wb} (*efficiency of the water boiling systems being replaced*) must be the weighted average based on the extent of usage of each boiling technology/device for drinking purposes per week. In other words, the **proportion of times a household uses a specific technology/device to boil water for drinking purposes over a**

week shall be applied as weights to the respective efficiencies of these boiling technologies for calculating η_{wb}^3 .

8.2 Project Emissions

Project emissions must be determined following the procedures provided in the most recent version of *AMS-III.AV*.

Paragraph 24 of *AMS-III.AV* must be replaced as follows:

- 24) $PE_{EC,y}$ must be calculated using the most recent version of *VT0010 Emissions from Electricity Consumption and Generation*. To estimate the amount of electricity consumption of water purification systems, the manufacturer's specification of the equipment (e.g., electricity consumption per hour times utilization hours, electricity consumption per liters of water treated) may be used.

8.3 Leakage Emissions

Projects must apply an adjustment factor to account for leakage related to the non-renewable woody biomass saved by the project activity (Adj_{LE}). For simplification, the adjustment factor is included in the calculation of baseline emissions in Equation (1). Leakage emissions must be determined ex ante following the procedures provided in the most recent version of *AMS-III.AV*.

8.4 Net Reductions and Removals

The net GHG emission reductions must be determined following the procedures provided in the most recent version of *AMS-III.AV*.

9 MONITORING

Project proponents must follow the monitoring procedures of the most recent version of *AMS-III.AV* and apply the changes indicated in this section.

9.1 Data and Parameters Available at Validation

Data/Parameter	EF_{b,k,CO_2}
Data unit	t CO ₂ /TJ

³ For instance, a household that boils water for drinking 10 times per week: 7 times on a traditional three-stone fire (device efficiency = 10%) and 3 times using an improved cookstove (device efficiency = 35%). The household-specific baseline efficiency (η_{wb}) is calculated as: $(0.7 \times 10\%) + (0.3 \times 35\%) = 17.5\%$.

Description	CO ₂ emission factor for fuel used by baseline device type <i>k</i> in the baseline scenario
Equations	(1)
Source of data	The following data sources may be used, listed in descending order of preference: - Option 1: Project-specific value - Option 2: Regional or national default values - Option 3: Default value from the most recent version of the <i>IPCC Guidelines for National Greenhouse Gas Inventories</i>
Value applied	Depends on the source of data chosen
Justification of choice of data or description of measurement methods and procedures applied	The values must be determined ex ante using one of the following sources: 1) Testing in accredited/recognized laboratories: Measurements must be undertaken in line with national or international fuel standards 2) Use of national default values 3) Use of default values from the most recent version of the <i>IPCC Guidelines for National Greenhouse Gas Inventories</i>. Values from 2006 are given below: Wood: 112 t CO₂/TJ Charcoal: 112 t CO₂/TJ (combustion only): Apply where renewable charcoal is used in the project and emissions from production of charcoal are estimated as project emissions ($PE_{fugitive,y}$) in line with VM0050. 165.22 t CO₂/TJ (combustion and charcoal production emissions): Apply where non-renewable biomass charcoal is used in the baseline, or in the baseline and project. IPCC guidelines are a recognized source.
Purpose of data	Calculation of baseline and project emissions
Comments	None

Data/Parameter	$EF_{b,k,nonCO2}$
Data unit	t CO ₂ e/TJ
Description	Non-CO ₂ emission factor for fuel used by baseline device type <i>k</i> in the baseline scenario

Equations	(1)
Source of data	The following data sources may be used, listed in descending order of preference: - Option 1: Project-specific value - Option 2: Regional or national default values - Option 3: Default value from the most recent version of the <i>IPCC Guidelines for National Greenhouse Gas Inventories</i>
Value applied	Depends on the source of data chosen
Justification of choice of data or description of measurement methods and procedures applied	The values must be determined ex ante using one of the following sources: 1) Testing in accredited/recognized laboratories: Measurements must be undertaken in line with national or international fuel standards 2) Use of national default values 3) Use of default values from the most recent version of the <i>IPCC Guidelines for National Greenhouse Gas Inventories</i>. Values from 2006 are given below: Wood: 9.46 t CO₂e/TJ (AR5 GWP) Charcoal: 5.865 t CO₂e/TJ (combustion only, AR5 GWP) 44.83 t CO₂e/TJ (includes charcoal production)
Purpose of data	Calculation of baseline emissions
Comments	None

Data/Parameter	Adj_{LE}
Data unit	fraction
Description	Adjustment factor to account for leakage related to non-renewable woody biomass saved by the project activity
Equations	(1)
Source of data	Procedures from the most recent version of <i>AMS-III.AV</i> . must be applied
Value applied	N/A
Justification of choice of data or description of measurement methods and procedures applied	Procedure from the most recent version of methodology <i>AMS-III.AV</i> .

Purpose of data	Calculation of baseline emissions
Comments	Parameter fixed at validation

9.2 Data and Parameters Monitored

Data/Parameter	$f_{NRB,i,y}$
Data unit	Fraction or %
Description	Fraction of non-renewable fuel type i used in the absence of the project activity in year y
Equations	(1)
Source of data	Where the baseline fuel is woody biomass, one of the following sources must be used: 1) Most recent national default values published by the UNFCCC 2) Calculated based on <i>CDM TOOL30</i> and applying an uncertainty discount of 26%⁴ 3) Default value from the most recent version of <i>CDM TOOL33</i>
Description of measurement methods and procedures to be applied	See "Source of data."
Frequency of monitoring/recording	Each crediting period. Project proponents applying draft UNFCCC national default values must apply the updated approved default values in any verification request submitted after UNFCCC approval of the default values.
QA/QC procedures to be applied	Where using UNFCCC f_{NRB} values, project proponents must report the related uncertainties along with the value in the project description.
Purpose of data	Calculation of baseline emissions
Calculation method	N/A
Comments	$f_{NRB,i,y}$ is equal to one where the baseline fuel is fossil fuel.

⁴ For example, if $f_{NRB,i,y}$ calculated based on *TOOL30* is 0.60, then the $f_{NRB,i,y}$ applied to GHG emission reduction calculations is $0.60 \times (1 - 0.26) = 0.44$. The 26% uncertainty discount is derived from 100% uncertainty in $f_{NRB,i,y}$, assuming a normal distribution. The discount is calculated using a standard deviation of 61% ($100\%/1.6449$) and a one-sided 66.6% confidence interval ($61\% \times 0.4307 = 26\%$). A fixed uncertainty discount is applied across all projects that use *TOOL30* to calculate $f_{NRB,i,y}$.

9.3 Description of the Monitoring Plan

All monitoring requirements of *AMS-III.AV.* and applied tools must be followed in addition to the following requirements:

- 1) Both the baseline and project survey questionnaires must contain a survey question that allows the project proponent to capture any instances of interactive effects occurring between a safe drinking water system/device and improved thermal energy generation technologies in a residence or institution. The survey results related to this aspect must be applied proportionally across the population of households to determine the emission reductions.
- 2) The project proponent must compile data on each distributed device that is derived from the total sales record, with project technologies differentiated by different project scenarios. These data must be differentiated into sections based on the results of the applicable monitoring studies for each project scenario, so that GHG emission reductions can be calculated appropriately, section by section. The information captured for each project device included in the project activity must include, at a minimum:
 - a) Date of sale
 - b) Geographic area of sale
 - c) Model/type of project technology sold/distributed
 - d) Name, telephone number (where available), and address of recipient
 - e) Unique alphanumeric ID or GPS coordinates of each device that is sold/distributed

10 REFERENCES

None

DOCUMENT HISTORY

Version	Date	Comment
v1.0	31 Oct 2025	Minor revision, including the following changes: • Introduction of an updated approach to determine the fraction of non-renewable biomass • Fixing at validation of the leakage adjustment factor to account for leakage related to the use of non-renewable woody biomass saved by the project activity • Emission factor for non-renewable woody biomass (for both CO₂ and non-CO₂ components) and fossil fuels • Requirement related to the assessment of double counting with REDD+ projects and jurisdictional REDD+ programs • Requirements related to adjusting the level of baseline emissions of the residence/institution included in the project to account for the effects of interacting technologies • Requirements related to the compilation and presentation of relevant data for each distributed device